Air Compressors

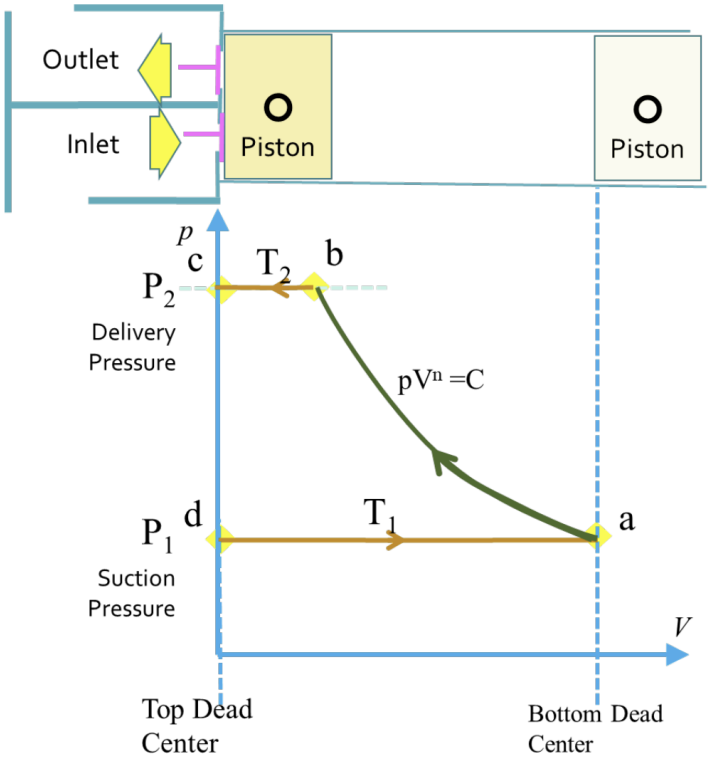
Generally, air compressors compress a low pressure gas into a higher pressure to be delivered elsewhere.

There are many different kinds of air compressors, but the main one covered is the reciprocating piston compressor, with a similar setup to gas powered engines.

The main difference is that work is done BY the piston, i.e. the crankshaft and piston is powered by an electric motor to yield a compressed gas.

Reciprocating Compressor 88without Clearance Volume

Consider a reciprocating compressor without any clearance volume. It undergoes the following cycle:



d → a: induction stroke (NOT ISOBARIC)

piston moves back → inlet valve opens and gas enters at const pressure and temp → piston reaches bottom dead center and value closes

a → b: polytrophic compression/ compression stroke

gas is polytrophically compressed to delivery pressure

b → c: delivery stroke (NOT ISOBARIC)

piston further moves as outlet valve openes, gas exits at const pressure and temp.

c → d: instantaneous pressure drop (not a real processs)

This also means that there are two distinct temperatures, T1 at a and d, T2 at b and c.

The work done on the gas (or the negative of the standard work done) for one cycle is

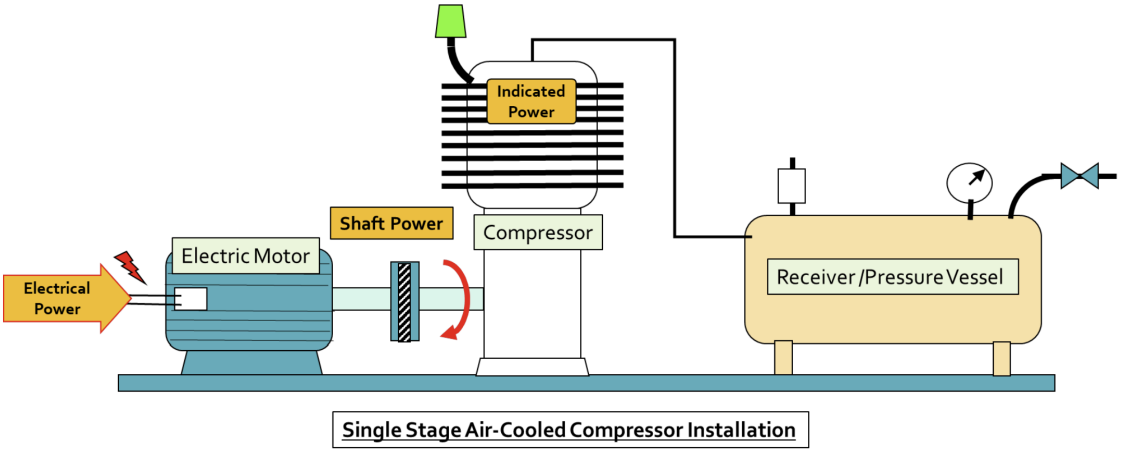
$$W = \frac{n}{n-1} mR(T_2 - T_1)$$

where T2 can be found either by

$$T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \frac{P_2 V_b}{mR}$$

However, the actual work done on the gas is more than this value, to account for the extra work needed to overcome friction.

Typically, the full setup involves a motor and a receiver vessel:



Ideally, the electrical power is equal to the shaft power, which is equal to the ideal power needed for the air compressor. However, friction causes

$$\text{Electrical Power} > \text{Shaft Power} > \text{Ideal, or Indicated Power}$$

Thus, the compressor efficiency

$$\eta_c = \frac{W_i}{W_s} = \frac{P_i}{P_s}$$

and the motor efficiency is

$$\eta_m = \frac{W_s}{W_e} = \frac{P_s}{P_e}$$

To minimise the work done, the gas should be compressed isothermally, as it minimises the area on the PV diagram, and hence the work done on the gas.

To achieve this minimal work done, n should be as close to 1 as possible, however n is usually between 1 to 1.4.

We can quantify how close the work done is to the minimum, aka the isothermal efficiency, where

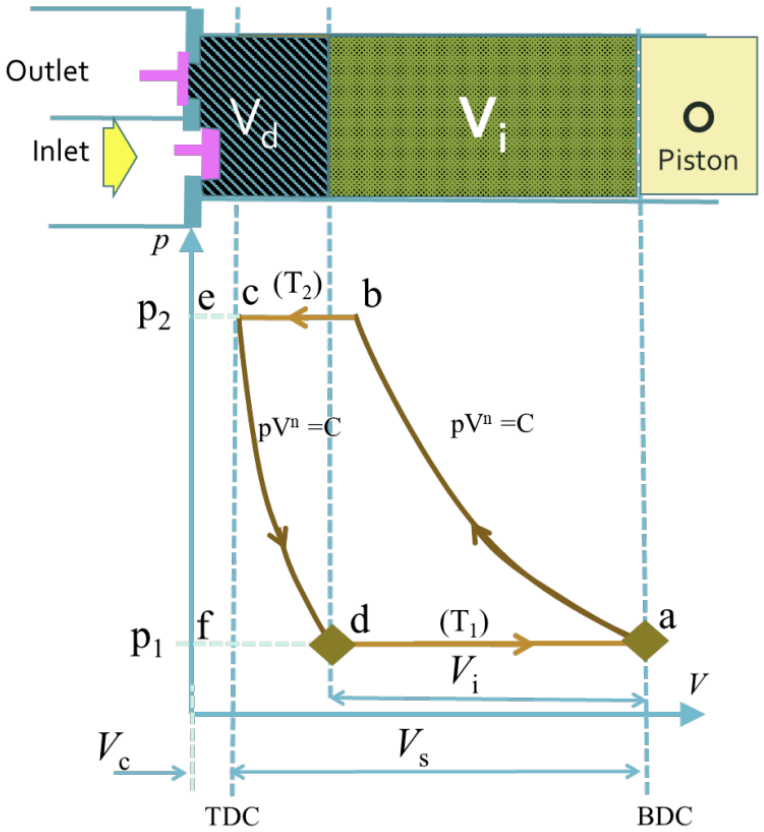
$$\text{Isothermal Work} = mRT_1 \ln\left(\frac{P_2}{P_1}\right)$$

and

$$\eta_{\text{isothermal}} = \frac{\text{isothermal work}}{\text{actual work}}$$

Reciprocating Compressor with Clearance Volume

Practically, air compressors have a small clearance volume for valve operations and piston movement:



c → d: polytrophic clearance expansion

The air starts at P2 and T2, and the piston causes an expansion process to let the remaining air reach P1 (and temp T1).

d → a: intake stroke

Air is then induced into the cylinder, as the inlet is opened to take in air at a constant temp and pressure.

a → b: polytrophic compression

The inlet closes and the air is compressed to P2 and T2.

b → c: delivery stroke

The outlet opens for the gas to exit at constant pressure and temp.

The induced volume, the volume span from d → a, can be found as

$$V_i = V_a - V_d = V_s + V_c - V_d$$

where Vs is the swept volume and Vc is the clearance volume.

Vd can be found with c → d:

$$P_d V_d^n = P_c V_c^n \Rightarrow V_d = V_c \left(\frac{P_c}{P_d}\right)^{\frac{1}{n}}$$

If the speed of the compressor is given as N in rpm, we can find the volume of air induced per second by converting rpm to revolutions per second:

$$\dot{V_i} = V_i \times \frac{N}{60}$$

With the clearance volume, the work done can be found similarly, with this formula:

$$W = \frac{n}{n-1}(m_a - m_d)R(T_2 - T_1)$$
$$W = \frac{n}{n-1}(\Delta m)R(T_2 - T_1)$$

The change in mass of the air is the same as the mass of the induced air, and since

$$P_1 V_i = (\Delta m)RT_1$$
$$\Delta m = \frac{P_1 V_i}{RT_1}$$

the work done (on the gas) is

$$W = \frac{n}{n-1} \times \frac{P_1 V_i}{\cancel{R}T_1} \times \cancel{R}(T_2 - T_1)$$
$$W = \frac{n}{n-1} \times (P_1 V_i) \times \frac{T_2 - T_1}{T_1}$$

T2 can also be found with polytropic equations, as

$$T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}$$

To find the (indicated) power, $\dot{W}$,

$$\dot{W} = \frac{n}{n-1}(\dot{\Delta m})R(T_2 - T_1)$$
$$\dot{W} = \frac{n}{n-1} \times (P_1 \dot{V_i}) \times \frac{T_2 - T_1}{T_1}$$

Free Air Delivery

To rate different air compressors, which may operate at different temperatures and pressures, we set them all to a common metric of using atmospheric conditions.

So, if an air compressor induces a certain volume of gas at P1 and T1, we can compare it to a certain volume of air being induced at atmospheric conditions. Hence, this can be expressed as

$$\frac{P_1 V_i}{T_1} = \frac{P_{atm} V_{FAD}}{T_{atm}}$$

A higher FAD volume means that it can pressurise more air/take in more air.

Volumetric Efficiency

The volumetric efficiency at the inlet can be found as

$$\eta_{v(inlet)} = \frac{V_i}{V_s}$$

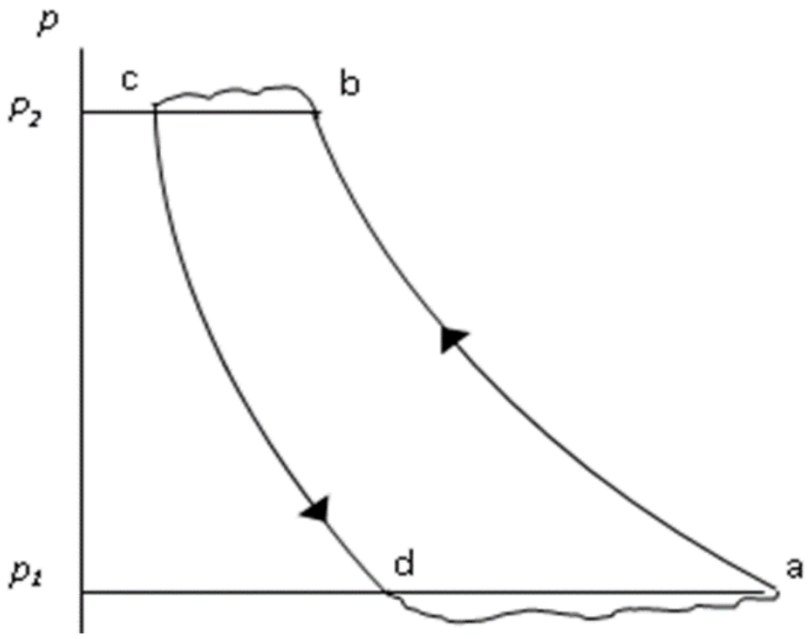
For air compressors without any clearance volume, $\eta_v = 1$.

We can also find the volumetric efficiency at free air conditions, as

$$\eta_{v(FAD)} = \frac{V_{FAD}}{V_s}$$

Indicator Diagram

The PV diagram used is an idealised version of the reciprocating air compressor. Instead, during the induction and delivery stroke, the (indicator) diagram will appear wavy.



During the initial start of d → a (induction), the valves have some inertia and do not open immediately once P1 is reached. Instead, the piston continues to move back as pressure drops further from P1 as the inlet takes time to open. Once it opens, the valve takes time to stabilise, and is shown by the valve bounce, or the wavy section of d → a.

Similarly for b → c, the pressure increases higher than P2 due outlet valve having some inertia, causing the air to be compressed slightly more as the valve takes time to open. The outlet also has some valve bounce, causing the wavy section of b → c.

With more valve bounce, more work is done on the gas than necessary (i think)